

KHONUU, Russian Federation

WMO: 243820

Lat: 66.450N

Lon: 143.233E

Elev: 196

StdP: 98.99

Time Zone: 11.00 (E11)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -51.3	(c) -49.6	(d) -52.7	(e) 0.0	(f) -50.9	(g) -51.1	(h) 0.0	(i) -49.5	(j) 6.4	(k) -23.2	(l) 5.2	(m) -21.9	(n) 0.0	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	28.7	17.4	26.1	16.0	23.7	15.2	18.6	26.7	17.3	24.4	16.0	22.1	2.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	11.3	20.9	14.3	10.4	19.8	13.1	9.6	18.8	53.6	27.0	49.5	24.7	45.7	22.4	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.8	6.0	5.2	-52.3	32.8	1.3	2.3	-53.2	34.5	-54.0	35.8	-54.7	37.1	-55.6	38.8
			-52.3	21.1	1.3	1.3	-53.2	22.0	-54.0	22.8	-54.7	23.5	-55.6	24.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.7	-40.0	-37.7	-25.0	-8.5	4.7	14.1	15.7	12.4	3.9	-12.2	-29.6	-39.2
	DBStd	21.76	6.65	6.49	7.82	6.88	5.43	4.10	4.41	4.08	4.67	7.19	7.37	5.78
	HDD10.0	8338	1549	1336	1086	554	179	11	4	23	191	689	1189	1527
	HDD18.3	10983	1807	1569	1344	804	422	136	108	188	434	948	1439	1785
	CDD10.0	436	0	0	0	0	16	134	181	97	8	0	0	0
	CDD18.3	39	0	0	0	0	0	9	26	3	0	0	0	0
	CDH23.3	603	0	0	0	0	4	152	366	82	0	0	0	0
	CDH26.7	187	0	0	0	0	1	30	141	16	0	0	0	0
Wind	WSAvg	1.5	0.3	0.3	1.3	2.1	2.7	2.6	2.3	2.2	2.1	1.6	0.5	0.3
Precipitation	PrecAvg	196	8	7	5	5	11	27	40	40	20	12	11	8
	PrecMax	273	22	21	17	14	34	69	87	106	57	30	32	16
	PrecMin	133	2	0	0	0	0	5	4	7	1	1	2	1
	PrecStd	38	4	5	4	3	9	14	19	24	13	8	6	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-19.9	-17.5	-1.3	8.7	23.5	30.1	34.3	29.1	21.3	4.3	-1.7	-20.9
		MCWB	-20.3	-18.2	-4.3	2.8	12.3	16.9	19.6	18.4	12.5	0.0	-4.3	-21.2
	2%	DB	-22.6	-21.8	-4.5	6.3	20.1	27.1	30.6	25.6	18.3	0.8	-12.9	-24.3
		MCWB	-22.8	-22.0	-7.0	1.4	10.7	15.3	18.5	16.1	10.5	-1.7	-13.6	-24.6
	5%	DB	-26.5	-24.8	-7.5	4.7	17.2	25.0	27.4	23.2	15.5	-1.0	-16.8	-28.0
		MCWB	-26.6	-25.1	-9.4	0.2	9.2	14.5	17.6	15.1	9.2	-2.4	-17.1	-28.2
	10%	DB	-29.8	-27.8	-11.0	2.7	14.6	22.8	24.5	20.9	12.3	-2.5	-19.1	-30.3
		MCWB	-29.9	-28.0	-12.4	-1.1	7.6	14.0	16.6	14.3	7.1	-3.7	-19.4	-30.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-20.2	-18.0	-4.1	3.2	12.5	18.0	20.9	19.2	13.0	0.7	-4.2	-21.2
		MCDB	-20.1	-17.5	-1.1	8.4	21.9	27.1	30.6	27.5	20.5	3.4	-1.5	-21.0
	2%	WB	-22.7	-22.0	-6.9	1.6	11.1	16.5	19.2	17.1	11.0	-1.0	-13.4	-24.5
		MCDB	-22.3	-21.7	-4.6	6.1	18.7	25.1	27.9	24.0	17.6	0.6	-12.9	-24.2
	5%	WB	-26.7	-25.1	-9.4	0.4	9.6	15.3	18.2	15.8	9.5	-2.3	-17.2	-28.1
		MCDB	-26.5	-24.9	-7.6	4.4	16.7	23.2	25.5	21.4	14.8	-1.0	-16.9	-28.0
	10%	WB	-29.9	-27.9	-12.3	-1.1	8.0	14.3	17.1	14.6	7.8	-3.7	-19.4	-30.3
		MCDB	-29.8	-27.7	-11.0	2.4	14.0	21.7	23.9	19.8	12.1	-2.5	-19.0	-30.2
Mean Daily Temperature Range	5% DB	MDBR	6.3	9.9	16.5	14.6	11.0	11.4	10.6	11.3	9.6	8.2	6.9	5.3
		MCDBR	9.3	10.8	17.1	13.4	14.9	15.6	17.0	16.9	16.1	8.4	9.2	9.0
		MCWBR	9.2	10.5	15.0	9.6	7.7	6.8	7.3	8.7	9.3	6.5	8.6	8.9
	5% WB	MCDBR	9.4	10.6	16.8	12.7	14.3	14.4	14.4	14.3	14.8	6.5	9.1	9.0
		MCWBR	9.2	10.3	14.8	9.2	7.3	7.0	6.8	7.7	8.8	5.0	8.6	8.9
Clear-Sky Solar Irradiance	taub		0.161	0.227	0.286	0.337	0.378	0.374	0.448	0.394	0.309	0.250	0.171	0.165
	taud		1.917	2.103	2.086	2.038	2.193	2.390	2.162	2.309	2.472	2.240	1.935	1.914
	Ebn at Noon		277	641	760	806	813	833	741	751	761	604	233	28
	Edh at Noon		15	52	91	125	120	102	125	97	64	45	14	3
All-Sky Solar Radiation	RadAvg		0.09	0.78	2.56	4.78	5.55	5.46	4.87	3.69	2.15	1.01	0.20	0.01
	RadStd		0.01	0.06	0.11	0.15	0.21	0.39	0.53	0.29	0.19	0.11	0.01	0.00

Historical Trends

		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (0 neighbors)		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page